Hot Hand, Real or Believed?

A behavioural analysis of setting decisions in professional volleyball — Short Research Paper

Studies on Human Behaviour — Course Project

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GitHub repository: https://github.com/FraMichi/Studies_on_human_behaviour.git

1. Introduction

Team sports provide useful observational data for studying human behaviour and decision-making in real competitive situations. In volleyball, scouting software such as Data Volley records each action of a match, making it possible to analyse both player performance and tactical decisions.

This study uses one season of matches from two Italian Serie A2 women's volleyball teams, Valsabbina Millenium Brescia and CDA Volley Talmassons FVG. Both teams were promoted to Serie A1 at the end of the season and were separated by only one point in the standings before the playoffs. The analysis focuses on the hot hand: the belief that a player who has just succeeded is more likely to succeed again. This phenomenon was first systematically studied by Gilovich, Vallone, and Tversky (1985), who found no evidence of a hot-hand effect in basketball shooting sequences.

The hot hand can be studied from two different perspectives. First, previous success may actually increase the probability of succeeding on the next attempt. Second, decision-makers may behave as if this effect exists even when performance does not change. Volleyball is particularly useful for separating these two aspects because the setter repeatedly decides which attacker receives the ball, and these decisions can be reconstructed from scouting data.

2. Research question

This project considers three different questions related to the hot hand:

RQ1 — Real hot hand. Does an attacker's probability of success depend on whether her previous attack in the same match was successful?

RQ2 — Believed hot hand. Is a successful attacker more likely to be selected again on the team's next attack, after accounting for differences between players, matches, and rotation?

RQ3 — Competitive pressure. Does the critical phase of a set affect attack-error probability or the use of high balls?

The two setters considered in the analysis (Vittoria Prandi from Brescia and Francesca Scola from Talmassons) played in the same league and season, allowing their allocation patterns to be compared in similar competitive contexts.

Comparing RQ1 and RQ2 makes it possible to examine whether allocation behaviour is consistent with the actual evidence of a hot-hand effect. In particular, a significant allocation pattern in RQ2 without a corresponding performance effect in RQ1 would suggest a difference between observed performance and the setter's decisions.

3. Methodology

3.1 Data and parsing

The dataset consists of 63 Data Volley (.dvw) scouting files from the 2025/2026 season of Brescia and Talmassons, covering the regular season, Coppa Italia, and the promotion play-offs.
The files were parsed using pydatavolley (OpenVolley), an open-source Python library that provides information about each scouted action, including team, player, skill, evaluation, attack-combination code, score, and player positions. Attack tempo was additionally decoded from the [3ATTACKCOMBINATION] section of the original files.
After cleaning, 8,312 attacks performed by 23 players from the two focal teams formed the analytical base.

3.2 Statistical approach

Observations from the same player and match are not independent. Some players have a higher baseline probability of success, while matches may also differ in difficulty. A simple comparison of raw proportions could therefore reflect these differences rather than the effect being studied.
For each research question, separate logistic mixed-effects models were estimated for Brescia and Talmassons, with crossed random intercepts for player and match:

$$logit\ P(outcome) = \beta_0 + \beta_{1(predictor)} + u_{\{player\}} + u_{\{match\}}$$

This allows the effect of the predictor to be estimated while accounting for baseline differences between players and matches.
The models were estimated in Python using an iterative procedure and tested on simulated data with known parameters before being applied to the volleyball data.

4. Analysis

For RQ1, each attack was linked to the outcome of the same player's previous attack in the same match, giving 7,861 usable observations. For RQ2, each attack was linked to whether the same attacker was selected again on the team's next attack, giving 8,247 observations.
Attack success was defined as a direct point (Data Volley evaluation code "#"), while an attack error was identified by the code "=".
For RQ3, an attack was classified as critical when at least one team had reached 20 points in sets 1–4, while the entire fifth set was considered critical. Attack tempo was decoded from the attack-combination code and classified as quick, super, medium, high, or other. Tempo was identified for all 8,312 attacks.
RQ2 also required a control for rotation. A team rotates when it wins a rally while receiving, meaning that previous success and the attacking options available on the next action can be related through the rules of the game. Rotation was identified by comparing the own team's setter position at the current and next team attack. After requiring this information for both actions, 8,182 observations were available for the rotation-controlled analysis.

5. Results

5.1 RQ1 — is there a real hot hand?

A comparison of raw proportions shows a higher success rate after a previous success for both teams: 40.2% vs. 36.8% for Brescia and 42.1% vs. 40.4% for Talmassons.
However, after accounting for differences between players and matches, neither effect is statistically significant (Brescia: $\beta = +0.085$, SE = 0.064; Talmassons: $\beta = +0.031$, SE = 0.071). Player-level variance is not negligible, particularly for Brescia ($\sigma^2 \approx 0.11$ on the log-odds scale), confirming that attackers differ in their baseline probability of scoring.
The differences observed in the raw proportions may therefore be partly explained by player heterogeneity. Players with higher baseline success rates are naturally more likely to appear in the "previous attack succeeded" condition, which can create an apparent pattern when observations are aggregated across players.
Overall, the mixed-effects models provide no statistically significant evidence of a real hot-hand effect in either team. This differs from Raab, Gula, and Gigerenzer (2012), who found streaks for about half of the elite volleyball players in their sample. Their study covered top first-division players and defined a streak over two or three consecutive attacks, while the present analysis considers second-division teams and only the immediately preceding attack, so the two results are not directly comparable.

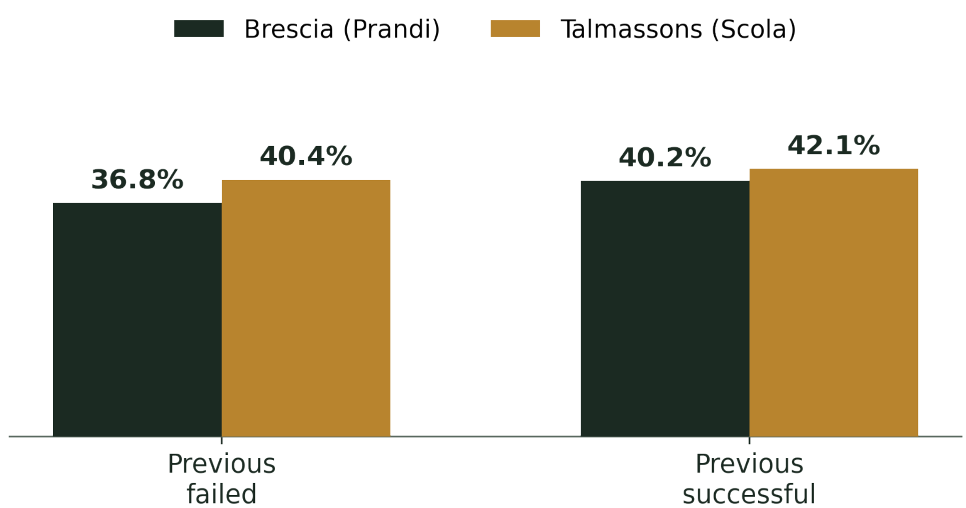

Figure 1. RQ1 — Attack success rate by the outcome of the same player's previous attack (raw rates). The difference observed for Brescia is not statistically significant in the mixed-effects model

5.2 RQ2 — Is setting behaviour consistent with a hot-hand belief?

The initial results suggest different allocation patterns between the two teams. For Talmassons, the same attacker is selected again after a success in 21.4% of cases, compared with 27.3% after a non-success. This difference remains statistically significant after accounting for player and match ($\beta = -0.295$, SE = 0.082, $p < .01$).
For Brescia, the probability of selecting the same attacker again is almost identical after a success and a non-success (31.8% vs. 31.7%; $\beta = +0.047$, SE = 0.067).

At this stage, the results would suggest an allocation pattern for Talmassons in the opposite direction to a hot-hand belief, while no clear pattern appears for Brescia. However, this model does not yet account for rotation.

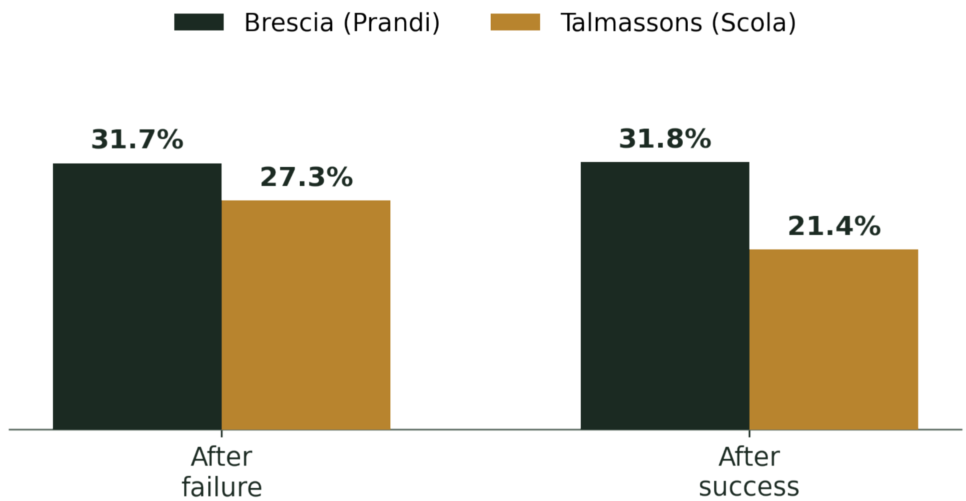

Figure 2. RQ2 — Rate at which the same player is selected again on the team's next attack, according to the outcome of her previous attack (raw rates).

5.3 RQ2 revisited — the rotation confound

The initial interpretation changes when rotation is considered. A successful attack is followed by a rotation in about 70% of cases (70.2% for Brescia and 69.3% for Talmassons), compared with about 15% after an unsuccessful attack (15.0% and 14.5%, respectively). When a rotation occurs, the probability of selecting the same attacker again also decreases, from 34.0% to 27.4% for Brescia and from 27.4% to 20.8% for Talmassons. Previous success and the attacking options available on the next action are therefore partly related through the rules of the game.

After adding rotation to the model, the RQ2 results change. For Talmassons, the negative effect becomes smaller and is no longer statistically significant ($\beta = -0.283$, $p < .01$ without rotation on the same sample; $\beta = -0.142$, SE = 0.098 with rotation). This suggests that the initial negative pattern was partly explained by rotation.

For Brescia, the opposite pattern appears. The effect is close to zero without the control ($\beta = +0.043$, not significant), but becomes positive and statistically significant after including rotation ($\beta = +0.246$, SE = 0.081, $p < .01$). Rotation itself has a negative coefficient for both teams (Brescia: $\beta = -0.380$, SE = 0.083; Talmassons: $\beta = -0.263$, SE = 0.100), confirming its relationship with a lower probability of selecting the same attacker again.

The revised RQ2 results therefore show no significant hot-hand allocation pattern for Talmassons. For Brescia, however, a successful attacker is more likely to be selected again after accounting for player, match, and rotation, which is consistent with a hot-hand allocation pattern.

5.4 RQ3 — does pressure change how the team attacks?

The interpretation of high-ball use requires some caution. High balls should not simply be considered a more conservative attacking option, because their use strongly depends on the quality of the previous

action. Faster attacks such as quick and super tempos generally require a good reception or defensive action, while a poor first contact makes a high ball more likely.
This relationship is clearly visible in the data. Among side-out attacks, the high-ball rate is 0.0% after a perfect reception (n = 887), compared with 59.8% after a poor reception (n = 914).
High-ball use should therefore not be interpreted as a direct measure of attacking caution. Instead, it can also be considered an indirect indicator of the quality of the build-up before the attack.

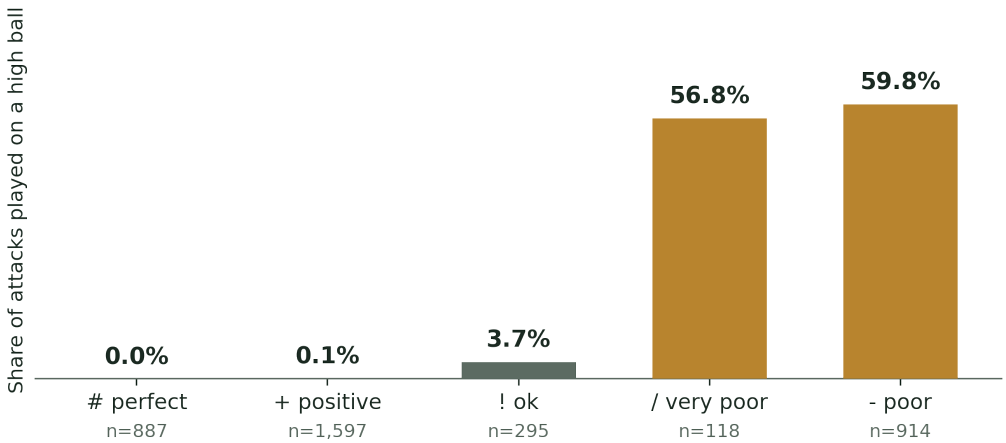

Figure 3. Share of attacks played on a high ball according to the Data Volley evaluation of the preceding reception. Side-out attacks only (3,811 of 8,312 attacks); the remaining attacks occur in transition after a dig or freeball.

Overall, 27.3% of own-team attacks (2,273 of 8,312) occurred during a critical phase. Talmassons showed no statistically significant change in either attack-error probability ($\beta = -0.046$, SE = 0.144) or high-ball use ($\beta = -0.112$, SE = 0.094). Brescia, instead, showed significantly fewer attack errors ($\beta = -0.358$, SE = 0.145, $p < .05$) and a lower probability of using a high ball ($\beta = -0.190$, SE = 0.078, $p < .05$) during critical phases. Considering the relationship between high-ball use and the quality of the preceding actions, the second result should not be interpreted as Brescia becoming more willing to take risks. Instead, the lower use of high balls is consistent with the team reaching the attack in better conditions during critical phases. Overall, there is no evidence of worse attacking performance under score pressure.
Table 1 summarises the mixed-effects model results across all three research questions.

Comparison	Brescia (Prandi)	Talmassons (Scola)	Pattern
RQ1 — success ~ prior success (β)	+0.085 (SE 0.064)	+0.031 (SE 0.071)	No significant hot-hand effect

RQ2 — same attacker selected ~ success (β)	+0.043 (SE 0.068)	−0.283*** (SE 0.082)	Before rotation control: negative pattern for Scola
RQ2 — same attacker selected ~ success + rotation (β)	+0.246*** (SE 0.081)	−0.142 (SE 0.098)	Positive allocation pattern for Prandi
RQ2 — effect of rotation itself (β)	−0.380*** (SE 0.083)	−0.263*** (SE 0.100)	Negative rotation effect in both teams
RQ3a — error ~ critical phase (β)	−0.358** (SE 0.145)	−0.046 (SE 0.144)	Fewer errors for Brescia; no significant change for Talmassons
RQ3b — high ball ~ critical phase (β)	−0.190** (SE 0.078)	−0.112 (SE 0.094)	Lower high-ball use for Brescia, no significant change for Talmassons

*Table 1. Mixed-effects model coefficients (β, log-odds scale) across the three research questions. The RQ2 models use the same sample ($n = 8,182$), allowing a direct comparison before and after controlling for rotation. Significance: * $p < .10$; ** $p < .05$; *** $p < .01$.*

6. Conclusion

The results show no statistically significant evidence of a real hot-hand effect for either team after accounting for differences between players and matches. Although the raw success rates suggest a possible pattern, part of this difference may be explained by player heterogeneity.

For RQ2, rotation substantially changes the interpretation of the setter's allocation behaviour. Before controlling for rotation, Talmassons shows a significant negative pattern while Brescia shows no effect. After including rotation, the Talmassons effect is no longer significant, while Brescia shows a positive and significant tendency to select a successful attacker again. This allocation pattern is consistent with a hot-hand belief, even though RQ1 provides no evidence of a corresponding performance effect.

RQ3 provides no evidence of worse attacking performance during critical phases. Talmassons shows no significant change, while Brescia makes fewer attack errors and uses fewer high balls. Since high-ball use is strongly related to the quality of the preceding actions, the lower high-ball probability should not simply be interpreted as a change in tactical risk.

Overall, the study shows the importance of considering the structure of the game when analysing behavioural patterns in observational sports data. Player heterogeneity, rotation, and the interpretation of technical variables all affect the conclusions that can be drawn from the observed patterns. In particular, the rotation analysis shows how a structural factor can change the interpretation of setting decisions and should therefore be considered before attributing these patterns to player or setter behaviour.

7. References

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